

General identities and native certificates

Cauchy-Binet, bounded source populations and marked Legendre generators

Arbitrary-dimensional weighted squared minors, kernel-checked distinct-image counts and numeric selections, conditional integral uniqueness, and replay-bound rational marked continuation.

7

reusable theorem declarations
audited

14

generated population
declarations checked

2

general Lean modules added

Focused native proofs and bounded executable certificates. The complete tracked archive preserves the separate flavor merge and the historical research receipts.

Source, corpus receipts and reproduction commands accompany the repository snapshot.

What this continuation closes

This 6 October 2026 edition follows the native arithmetic and elliptic-fibre work and preserves the separate session's flavor merge. It proves arbitrary-dimensional ordered Cauchy-Binet and its weighted Gram specialization in Lean, connects bounded Boolean polynomial sources directly to generated kernel proofs of distinct-image count and selection, and formalizes uniqueness of integral recognition. The period work adds replay-bound marked continuation evidence for the second Legendre generator and inverse orientation. This is a complete tracked-tree archive with a monograph; the full research program remains broader than these closures.

The reproducible entry points are `make connected-closures-lean`, `make connected-closures-receipts`, `make nonflavor-frontier-lean` and `make test`. This continuation audits seven reusable theorem declarations and fourteen generated population declarations. Its namespace `lint` covers the two new Lean modules and the extended finite-domain module. All audited declarations use only the standard axioms `propext`, `Classical.choice` and `Quot.sound`. The focused targets do not refresh the historical whole-repository theorem census. The validation receipt records the exact tests run and binds current source hashes.

Arbitrary-dimensional Cauchy-Binet

For finite index sets n and e and matrices M of shape $n \times e$ and N of shape $e \times n$, Lean proves

$$(\text{card } n)! \det(MN) = \sum \text{over } p:n \rightarrow e \text{ of } \det(M[:,p]) \det(N[p,:]).$$

The proof expands the second minor over permutations, interchanges finite sums and reindexes selections by each permutation. The sign from permuting the selected columns matches the sign in the determinant expansion. Each resulting inner sum is the rectangular determinant expansion proved in the preceding push. The factorial is exactly the number of permutations. This argument works in any commutative ring, including positive characteristic, because it never divides by the factorial. Repeated selections contribute zero through repeated columns or rows, so degeneracies and zero weights need no exclusions.

The weighted Gram specialization takes an arbitrary $e \times n$ matrix B and weights w in a commutative star ring. Its Gram entries are sums of $\text{star}(B_{ia}) w_i B_{ib}$, and Lean proves

$$\begin{aligned} (\text{card } n)! \det \text{Gram}(B,w) \\ = \sum \text{over } p:n \rightarrow e \text{ of } (\text{product over } i \text{ of } w_i(p_i)) \text{star}(\det B_p) \det B_p. \end{aligned}$$

Conjugate transpose gives the starred determinant and row scaling gives the weight product. This is the full general weighted squared-minor identity with ordered selections, removing the old two-column limitation. In a characteristic-zero ring without zero divisors, vanishing of every maximal minor now proves the weighted Gram determinant zero by cancelling the nonzero factorial. No positivity or connected-support hypothesis is smuggled into this implication. A separate unordered-subset formulation would remove

the explicit multiplicity; general gain-graph rank, matroid greedy correctness and implementation refinement of the algebraic-bit sampler still need their own proofs.

Source-complete bounded population certificates

The new `boundedImage` native transcript is the image of a decidable source predicate restricted to the literal integer interval $[lo, hi]$. Its completeness theorem identifies output membership with an existential original parameter satisfying both bounds, the original predicate and the exact image equation. This supplies the source equivalence directly rather than accepting a producer's claimed equivalence. Selected-value uniqueness then follows from checked membership and zero-based distinct numeric rank.

`population_certificate` translates the existing structured and/or/not polynomial predicate language, including integer remainder comparisons, into fixed Lean arithmetic syntax. It retains the bounded source and arbitrary integer polynomial image, computes an expected deduplicated image count and requested numeric selections, and emits count, membership and rank proofs by `decide +kernel`. Compiling the emitted source forces the kernel to reduce the native finite-set evaluator and validate those claims. The Python calculation is only a producer. A JSON packet or emitted source that has not been compiled remains marked execution-unverified. Bounds are explicit semantics, not a purported proof that every unbounded source solution lies inside them.

The retained fixture filters -12 through 12 by $n \bmod 4 < 2$ or $n^2=9$ and projects n^2 . It has ten distinct values, and Lean certifies ranks $0, 3$ and 9 as values $0, 16$ and 144 . Three additional emitted fixtures check reversed empty bounds, constant-image deduplication, and nested negation/disjunction with negative integer remainders. Its completeness theorem still refers to the original accepted integer, so duplicate square values disappear without losing the source relation. This numeric output order differs from the existing `ProjectedPopulation` interface's increasing canonical-owner order; both are stated explicitly. The certificate path enforces integer, degree, source-width, Boolean-node and selection budgets and returns no partial completeness claim. It is deliberately an affordable bounded certificate route. Generic nonenumerating Sturm transcripts, semilinear cell emission, parser-to-native refinement and verified large rank/select algorithms remain open.

Integral recognition and marked continuation

`IntegralRecognition.integer_unique` proves that two integer candidates lying within a rational radius strictly below one half of the same rational centre are equal. Its matrix theorem applies entrywise in arbitrary dimensions. The proof uses the triangle inequality and the fact that an integer with absolute value below one is zero. Integrality and enclosure remain hypotheses. The analytic producer's complex row norm implies the required real-entry bounds, but that analytic-to-native connection, the Neumann inverse estimate and the Legendre cycle marking are not formalized by these two uniqueness theorems.

The new monodromy replay operation receives the path externally, recomputes rational Taylor disks, outward-rounded transport, seed errors, inverse and product balls, integral recognition and the symplectic check, and compares every serialized field. Canonical JSON comparison distinguishes true from 1 and binds markings, bounds and intermediate disks. It is a deterministic replay checker using the same analytic implementation, not an independent proof of its estimates. Tests alter matrix entries, bounds, seed metadata and marking text and require rejection; an open or changed path cannot borrow a closed-loop receipt.

The retained first generator around zero is $[[1,2],[0,1]]$. The new second-generator and reverse-orientation calculations use the same standard a,b marking and rational seed at $1/2$. The second loop yields $[[1,0],[-2,1]]$ in 148 certified steps with a row enclosure approximately $2.22466826496 \times 10^{-9}$. Reversing the zero loop gives $[[1,-2],[0,1]]$ in 148 certified steps, and exact integer multiplication verifies the inverse identity. The two generator products are $[[-3,2],[-2,1]]$ and $[[1,2],[-2,-3]]$, so the representation is noncommuting. These are algebraic products of individually certified matrices; this edition does not claim a fresh concatenated-path enclosure. The exact matrices and disk counts are retained alongside the complete path and settings. The pair provides a concrete noncommuting marked representation, and inverse orientation checks the sign convention. Arbitrary period markings, singular endpoints, higher-genus integral cycle transport and analytic estimates in Lean remain separate research obligations.

The connected repository context

Native roots become actual elliptic fibres

NativeRationalRoots turns the proved signed-divisor integer generator into a complete rational-root generator. A producer supplies an ascending integral coefficient list, a nonzero rational scale D , monicity and an exact coefficient identity with the scaled polynomial. Lean checks those premises. The integer-root output is obtained from NativePolynomialRoots; it is no longer supplied as a separately assumed complete list. Roots are transported back by z/D . A leading-coefficient normalization extends the result to every nonzero rational polynomial for which the checked scaled integral coefficients are supplied. Zero constant terms and repeated roots retain the existing recursive treatment.

EllipticPointDivision connects the completed-square model $y^2 = x^3 + Ax^2 + Bx + C$ to Mathlib's actual affine point group. For nonsingular affine points and nonzero 2 , two-torsion is equivalent to $y=0$; the x coordinate is therefore a cubic root. The x coordinate in Mathlib's doubling formula satisfies exactly the halving quartic proved in the previous push. A nonzero discriminant allows each rational cubic root to construct a nonsingular branch point. The torsion list includes infinity. Native rational roots then prove this list complete, and an actual checked anchor H with $2H=P$ gives all halves by translating that torsion list. The resulting theorem covers the entire fibre, including infinity and exceptional points through the group theorem; only the coordinate doubling equivalence uses its explicit nonzero-denominator premise.

The remaining elliptic obligations are now sharply located. The Python point-law implementation, JSON parser and certificate interpreter still need a formal refinement to these native definitions. Anchor discovery remains a producer task; its group equation is a checked premise, not a rank assertion. The independence machinery still proves the stated lower bounds under its own certificate semantics; full Mordell-Weil rank determination, arbitrary n -division and generic higher-genus covering maps remain separate work. Generalized-model transport must continue to preserve original-coordinate integrality.

Compositional domains, CRT and nonmonotone images

`FiniteDomainCertificate` provides a transcript language for literal finite sets, unions, intersections, decidable filters and images. Its denotation retains the source predicate, and its evaluator produces a deduplicated finite set. A structural completeness theorem relates evaluation to denotation. Rank counts distinct accepted integers strictly below a value. Strict rank growth and selected-value uniqueness show that membership plus rank identifies one value. A source-equivalence theorem transports that guarantee back to a supplied original predicate. This is useful for nonlinear images because several source parameters can have one output value; counting source occurrences would give the wrong rank.

`GeneralCRT` removes a hidden coprimality restriction from the mathematical chart layer. A simultaneous pair of natural residue conditions exists exactly when the residues agree modulo the gcd. The canonical compatible anchor has an lcm-period chart and, for nonzero moduli, lies below that lcm. A separate integer theorem keeps the same chart exact for negative parameters and residues. The remaining compiler work is a universal parser/AST correspondence, generic Sturm-variation certificates, assembly of every emitted semilinear cell and a verified algorithm for rank/select and optimizer search. The new transcript theorem proves the meaning of assembled checked data; it does not claim that an arbitrary Python compilation already constructs those proofs.

The population implementation now supports tuple-key joins. Bounded value joins preserve every original pair and every multiplicity. Symbolic joins choose one complete curve relation and enforce the additional polynomial equalities through the curve predicate language. A test joins the square and linear fields over $2 \times 10^{30} + 1$ source parameters and retains exactly the diagonal without enumeration. Unsupported primary curve relations still fail explicitly. Arbitrary multivariate joins and a general complete relation solver remain open.

Nonmonotone projection now includes integer even quartics $ax^4 + bx^2 + c$. `QuarticCollision` proves that equal values have $x=y$, $x=-y$ or $a(x^2 + y^2) + b = 0$, and proves bounds for the circle branch. The Python backend handles the infinite reflection through source predicates and exhausts the finite circle when its work bound permits. It retains the least original accepted owner even on disconnected domains, then reuses the existing image/rank interfaces. Arbitrary nonsymmetric quartics and higher-degree nonmonotone collision curves still require new complete backends; a budget rejection returns no partial completeness claim.

Variable-width factorial arithmetic

FactorialWindow proves the p-adic valuation of $N!/(N-W)!$ as a difference of factorial valuations and then as two finite Legendre sums. At the native level, natural subtraction makes $W > N$ a truncated window; the public Python operation requires $0 \leq W \leq N$. A valuation not divisible by k proves that the positive window product is not a k -th power. The Python witness search uses the bit length of N and a bounded supplied prime list, so it can process widths that cannot be enumerated. Passing every tested prime produces UNKNOWN.

The retained generated theorem proves that $(10^{30})!/(10^{30}-10^{20})!$ is not a square. Its 2-adic valuation is 100000000000000000011, which is odd. Lean evaluates the finite floor sums and checks logarithmic truncation bounds through a power inequality; it does not reduce either enormous factorial. This closes a useful variable-width Gamma-ratio obstruction route. Complete classification of arbitrary variable-width products, sums or differences of factorials, general Gamma arguments and multivariate creative telescoping remain open. Sequence-definition review, continued-fraction execution at very large unit height, broad nonlinear recurrences, transform endpoints and Tauberian conclusions retain their previous boundaries.

Certified periods, inverses and marked monodromy

The analytic extension uses rational complex centres and an induced row norm with $|\operatorname{Re} z| + |\operatorname{Im} z|$ on entries. For a supplied enclosure $\|A-C\| \leq e$, exact Gaussian elimination computes C^{-1} . If $q = \|C^{-1}\|e < 1$, Neumann's lemma proves nonsingularity throughout the ball and bounds inverse error by $\|C^{-1}\|q/(1-q)$. Product balls use the full cross term. Integral recognition requires error strictly below one half, checks entry and row compatibility, and can check the standard symplectic form exactly. The general operation explicitly requires an independent integrality premise: closeness alone cannot establish integrality.

Long analytic paths now have an optional outward-rounded dyadic mode. Centres are rounded with a bounded row error added to the enclosure; the error bound itself is rounded upward. Certified disks also keep the exponential bound controlled. This prevents repeated rational matrix products from accumulating huge denominators while preserving the existing exact mode. The marked Legendre wrapper combines the original certified seed, closed continuation, inverse and product balls, and symplectic recognition. Its integrality justification is the supplied standard integral cycle marking, not numerical proximity.

The retained nontrivial loop starts at $1/2$, passes through $1/2+i/4$, $-1/4+i/4$, $-1/4-i/4$ and $1/2-i/4$, and returns to $1/2$. At Taylor order 24 and 128-bit outward rounding, 148 certified steps identify the cycle matrix $[[1,2],[0,1]]$. The row enclosure is approximately $2.22466826496 \times 10^{-9}$. This is a concrete certified marked monodromy computation. General singular endpoint asymptotics, arbitrary integral period markings, rigorous quadrature with arbitrary cycle inputs, holomorphic-basis normalization and the analytic estimates themselves in Lean remain open.

Curve structure, arithmetic reduction and global geometry

The existing exact curve engine retains its explicit quotient towers, two-isogenies, Richelot constructions, de Rham connections, superelliptic reductions, horizontal-section searches and selected Frobenius computations. The current push adds native elliptic fibres and a certified marked analytic continuation bridge. General higher-genus correspondences still need integral cycle maps, kernel degrees and singular-target treatment. Nonsplit stable reduction, arbitrary Puiseux root jets, arithmetic component twists, finite-field extension descent, cross-parameter p-adic Frobenius transport, full differential Galois groups, nonlinear idempotents and conjugate Hodge/Betti descent remain research fronts. A bounded ansatz search for horizontal sections cannot prove that every possible section has been found.

`cluster_graph_metric` turns the existing split-root geometric semistable graph into a global finite length metric and tropical cycle lattice. Exact Floyd-Warshall distances respect all graph identifications and parallel edges. A deterministic spanning tree gives integral fundamental cycles; every boundary is checked to vanish, the number of cycles matches the graph's cycle rank, and the edge-length Gram form has a positive determinant. A replay operation binds the entire packet to the original rational roots and odd prime and rejects changed source data or witnesses. These distances and cycle lengths are properties of the geometric reduction graph. They are not a complex-curve metric, a minimal arithmetic regular model or a Frobenius marking.

The remaining smooth geometry work includes a global quotient length metric on the analytic atlas, JSON-to-native geometric checker correspondence, global chart gluing and smoothness comparison, general period-normalized metrics and certified geodesics. The existing 108 closed metric boxes out of 144 leave 36 unresolved boxes; this push does not turn them into certified boxes. Numeric hyperelliptic periods remain scoped to their supported genera and numerical methods, and general cyclic-mesh cycle integration still needs a marked analytic bridge.

Solver transport and verification consumers

`iter_commands` accepts bounded text chunks and yields exact SMT-LIB command slices on demand. Its state handles comments, quoted symbols and doubled string quotes across every chunk boundary. It rejects malformed tails, top-level garbage and exceeded command or chunk limits. `solve_replay_stream` preserves native query barriers and batches only non-query transport. A query sink avoids retaining rows. Errors stop future queries while preserving completed earlier query results. Storage bounds cover the scanner and pending transport; they do not bound the solver's assertion state, the caller's input storage or an intentionally retained result list.

The scanner matched 277345 exact slices across 292 repository SMT files at a 127-character chunk size. A source-hashed replay of 16 repository and Why3 tasks reported no sat/unsat contradictions. Scanner allocation measurements used generated chunks and measured 1332 bytes for 1000 commands and 1340 bytes for 100000 commands. They are local measurements, not whole-process memory bounds. The earlier

external 181-file industrial cohort is not replicated by this receipt. Its reported throughput and solved-query gains retain their earlier provenance, and independent replication and broader throughput work remain necessary.

The Why3 bridge now allocates distinct names after sanitization, reserves generated local names and language keywords, safely encodes source-symbol comments, caps expanded powers and rejects unsupported free Boolean binders. These fixes eliminate concrete malformed or capturing translations. The original Why3 tool is unavailable in this environment, so its native rerun remains skipped. A universal SMT-to-Why3 semantic correspondence, SPARK/GNATprove VC correspondence, overflow and bitwise refinements remain open. Brainpool's residue-ring equation/export transport remains proved, while its 384-bit field primality and complete group implementation bridge still require a native prime certificate; trial factoring of $p-1$ leaves a composite 349-bit cofactor.

Global arithmetic and next research decisions

The historical descent coverage still records 79 classes: 15 local and 64 conditional, with zero unresolved coverage entries. Unit registration covers every currently needed obligation; this does not discharge the global Matveev-type premises. The 23 negative curves carrying those premises retain conditional theorem statements. The positive registry still records 73 closed cases and the earlier conditional entries. $k=22$ remains closed by its retained 100-chunk kernel check of 971659 source candidates. $k=94$ remains the next larger source with 5294219 candidates; no complete source theorem for it is claimed by this push. Generic native maximal-order computation, basis-independent nonmonic norm transport and recentered Skolem extraction remain separate from the rational-polynomial normalization delivered here.

Density-zero statements for general nonrigid branches, normalized Kummer/Riemann-Hurwitz statements and global Siegel/Boshernitzan ingredients remain mathematical obligations. Hall, Pillai and uniform integral-point conjectures remain outside the unconditional results. The bounded finite-domain emitter and general ordered weighted identity now have native closures. The next productive work is to refine elliptic execution and nonenumerating domain compilation, package the unordered-subset determinant form, extend certified marked periods to singular and higher-genus settings, and choose an arithmetic closure with an affordable source-proof budget. No roadmap item is promoted to a theorem merely because it now has a working data pipeline.